Conformal Inference Assignment 2
(Deadline: October 18)

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Remark All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

Exercise 1

Consider split conformal prediction with data set $(X_1, Y_1), \cdots, (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$ and the conformal risk control with given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. Instead of what is in Chapter 10 of the book, we define, for all $\lambda \geqslant 0$:

$$\mathcal{C}_\lambda(x) = \{y \in \mathcal{Y} \mid s(x, y) < \lambda\}$$

and

$$L(y, \mathcal{C}_\lambda(x)) = \chi_{\{y \notin \mathcal{C}_\lambda(x)\}}.$$

Show that, with this definition, $L(y, \mathcal{C}_\lambda(x))$ is no longer always right continuous for every fixed $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Next, given $\alpha \in (0, 1)$, give a concrete expression for

$$\hat{\lambda} = \inf \left\{ \lambda \geqslant 0 \mid \hat{R}(\lambda) \leqslant \alpha - \frac{\alpha - 1}{n} \right\}$$

in terms of the data, where

$$\hat{R}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} L(Y_i, \mathcal{C}_\lambda(X_i)).$$

Show that, if we assume that $(S_1, \ldots, S_{n+1})$ is exchangeable and that the S_i's almost surely has no tie, then

$$P\left\{Y_{n+1} \in \mathcal{C}_{\hat{\lambda}}(X_{n+1})\right\} \geqslant 1 - \alpha.$$

Provide a counterexample to show that this need not hold if there are ties in S_i's with positive probability, even if they are exchangeable.

Solution Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$ be given. If $\{\lambda_m\}_{m=1}^{\infty}$ is any decreasing sequence such that $\lambda_m > s(x, y)$ for every m and $\lambda_m \to s(x, y)$ as $m \to \infty$, then $L(y, \mathcal{C}_{\lambda_m}(x)) = 1$ for all m but $L(y, \mathcal{C}_{s(x,y)}(x)) = 0$, proving that $L(y, \mathcal{C}_\lambda(x))$ is not always right continuous.

We move to compute $\hat{\lambda}$. The definition of $\hat{R}(\lambda)$ and simple arithmetic rearrangement shows that, for any $\lambda \geqslant 0$,

$$\hat{R}(\lambda) \leqslant \alpha - \frac{\alpha - 1}{n}$$

if and only if

$$\sum_{i=1}^{n} \chi_{\{S_i < \lambda\}} \geqslant (n + 1)(1 - \alpha).$$

Thus, it is obvious that

$$\hat{\lambda} = S_{(\lceil (n+1)(1-\alpha) \rceil)}.$$

Here we have used order statistics $S_{(1)} \leqslant \cdots \leqslant S_{(n)}$ for the first n values of $S_1, \ldots, S_n$. Consequently:

$$C_{\hat{\lambda}}(X_{n+1}) = \left\{y \in \mathcal{Y} \mid s(X_{n+1}, y) < S_{(\lceil (n+1)(1-\alpha) \rceil)}\right\}.$$

To prove coverage, let $\alpha \in (0,1)$ be given. Assume first that that $(S_1, \ldots, S_{n+1})$ is exchangeable and there is almost surely no tie in the S_i's. Then

$$P\left\{Y_{n+1} \in C_{\hat{\lambda}}(X_{n+1})\right\} = P\left\{S_{n+1} < S_{(\lceil (n+1)(1-\alpha) \rceil)}\right\} = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geqslant 1-\alpha,$$

where the second equality uses both assumptions: when $S_{(1)} < \cdots < S_{(n)}$, inserting a new exchangeable point S_{n+1} to get exactly right before $\lceil (n+1)(1-\alpha) \rceil$th place has the probability $\frac{\lceil (n+1)(1-\alpha) \rceil}{n+1}$. If we remove the assumption that there are no ties, then everything can go wrong. For example, just setting $s \equiv 0$. Then $\hat{\lambda} = 0$ and $C_{\hat{\lambda}}(x) = \emptyset$ for all x, clearly not yielding any good coverage probability. ////

Exercise 2

Consider split conformal prediction with data set $(X_1, Y_1), \cdots, (X_{n+2}, Y_{n+2}) \in \mathcal{X} \times \mathcal{Y}$ with given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. For $j \in \{1, 2\}$, the conformal p-value is the random variable defined by

$$p_j = \frac{1 + \sum_{i=1}^{n} \chi_{\{S_{n+j} \leqslant S_i\}}}{n+1}.$$

Given $\alpha \in (0,1)$, let $\mathcal{C}$ be the conformal interval calibrated only on the first n observations. In other words, for $j \in \{1, 2\}$, the event $\{Y_{n+j} \in \mathcal{C}(X_{n+j})\}$ equals to $\{p_j < \alpha\}$. Assuming that $((X_1, Y_1), \cdots, (X_{n+2}, Y_{n+2}))$ is exchangeable, s is a symmetric score function, and that the random vector (p_1, p_2) satisfies the positive regression dependence on subsets (PRDS) assumption given in definition 10.3 of the textbook, prove that

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1}),\, Y_{n+2} \in \mathcal{C}(X_{n+2})\} \geqslant (1-\alpha)^2$$

without explicitly computing the probability on the left hand side of the inequality.

Solution First of all, Theorem 3.2 in the textbook already guarantees that, for $j \in \{1, 2\}$:

$$P\{Y_{n+j} \in \mathcal{C}(X_{n+j})\} \geqslant 1-\alpha.$$

We focus on showing that

$$P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \geqslant 1-\alpha.$$

In order to make use of the PRDS condition, define a function g on the support of the discrete random variable p_1 by setting, for all t, that

$$g(t) = P\left(\{p_2 > \alpha\} \mid p_1 = t\right).$$

Note the following facts:

1. g is monotonically increasing because $g(t) = P\left(\{(p_1, p_2) \in A\} \mid p_1 = t\right)$ with the set A defined by all $(u, v) \in \mathbb{R}^2$ with $v < \alpha$, who is an increasing set.

2. The average value of $g(p_1)$ on the set $\{p_1 > \alpha\}$ is the desired conditional probability. In symbols:

$$\frac{1}{P\{p_1 > \alpha\}} \mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\}.$$

3. The expectation of $g(p_1)$ is the marginal covergae of (X_{n+2}, Y_{n+2}). In symbols:

$$\mathbb{E}\left[g(p_1)\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2})\right\}.$$

4. The average value of $g(p_1)$ on the set $\{p_1 > \alpha\}$ is greater or equal to the average value of $g(p_1)$ on the set $\{p_1 \leqslant \alpha\}$ because g is monotonically increasing. That is:

$$\frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] \geqslant \frac{1}{P\{p_1 \leqslant \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 \leqslant \alpha\}\right]$$

The proof is complete once we combine the last three facts with the marginal coverage on (X_{n+2}, Y_{n+2}):

$$\begin{aligned}
&1 - \alpha \geqslant P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2})\right\} = \mathbb{E}\left[g(p_1)\right] \\
&= \frac{P\{p_1 > \alpha\}}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] + \frac{P\{p_1 \leqslant \alpha\}}{P\{p_1 \leqslant \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 \leqslant \alpha\}\right] \\
&\geqslant (P\{p_1 > \alpha\} + P\{p_1 \leqslant \alpha\})\frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] \\
&= \frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\}.
\end{aligned}$$

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Exercise 3

Suppose $(X_1, Y_1), \ldots, (X_{n+2}, Y_{n+2}) \in \mathcal{X} \times \mathcal{Y}$ are exchangeable. After observing X_{n+1} and X_{n+2}, select $I \in \{1, 2\}$, where I may depend on both (X_{n+1}, X_{n+2}). Given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, define a (full) conformal prediction set $\mathcal{C}(X_{n+I})$ such that

$$P\{Y_{n+I} \in \mathcal{C}(X_{n+I})\} \geqslant 1 - \alpha.$$

Solution The solution is to use a Benferroni correction for the two test points.

Specifically, put $S_i = s(X_i, Y_i)$ and let $S_{(1)} \leqslant \cdots \leqslant S_{(n)}$ be the order statistics for $S_1, \ldots, S_n$. Define

$$k = \left\lceil (n+1)\left(1 - \frac{\alpha}{2}\right)\right\rceil,$$

$$\hat{q} = \begin{cases} S_{(k)}, & k \leqslant n, \\ \infty, & k = n+1, \end{cases}$$

and

$$\mathcal{C}(X_{n+I}) = \left\{y \in \mathcal{Y} \mid s(X_{n+I}, y) \leqslant \hat{q}\right\}.$$

Standard results (for example, for full conformal prediction, Theorem 3.2 in the textbook) show that, for $i \in \{1, 2\}$, we have

$$P\left\{Y_{n+i} \notin \mathcal{C}(X_{n+i})\right\} \leqslant \frac{\alpha}{2}.$$

Thus:

$$P\left\{Y_{n+I} \notin \mathcal{C}(X_{n+I})\right\} = P\left(\bigcup_{i=1}^{2}\left\{Y_{n+i} \notin \mathcal{C}(X_{n+i})\right\}\right) \leqslant 2 \times \frac{\alpha}{2} = \alpha.$$

This completes the proof. ////

Exercise 4

Consider split conformal prediction with calibration data set $(X_1, Y_1), \cdots, (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$ at given level $\alpha \in (0, 1)$ and given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. We look at the aggregating prediction sets by the majority vote $\mathcal{C}^{\text{mv}}(X_{n+1})$ of two conformal algorithms $\mathcal{C}_1$ and $\mathcal{C}_2$, where $\mathcal{C}_1$ uses s as score and $\mathcal{C}_2$ uses $-s$ as score. Give an explicit formula of $\mathcal{C}^{\text{mv}}(X_{n+1})$ and its marginal coverage level, assuming exchangeability of the data set, symmetry of s, and almost surely no ties in S_i's.

Solution We first compute the expression of each conformal set. Let us use the order statistics $S_{(1)} < \cdots < S_{(n)}$ of $S_1, \cdots, S_n$. Put

$$k = \lceil (n+1)(1-\alpha) \rceil.$$

We discuss by cases of whether $k = n + 1$.

First of all, in the usual case, $k < n + 1$. The rule $(-v)_{(i)} = -v_{(n-i+1)}$ for any $v \in \mathbb{R}^n$ tells us that, by definition of $\mathcal{C}_1$ and $\mathcal{C}_2$:

$$\mathcal{C}_1(x) = \left\{ y \in \mathcal{Y} \mid s(x, y) \leqslant S_{(k)} \right\}$$

and

$$\mathcal{C}_2(x) = \left\{ y \in \mathcal{Y} \mid s(x, y) \geqslant S_{(n+1-k)} \right\}.$$

Thus, by definition:

$$\mathcal{C}^{\text{mv}}(x) = \left\{ y \in \mathcal{Y} \mid \frac{1}{2} \sum_{i=1}^{2} \chi_{y \in C_i(x)} > \frac{1}{2} \right\} = \mathcal{C}_1(x) \cap \mathcal{C}_2(x) = \left\{ y \in \mathcal{Y} \mid S_{(n+1-k)} \leqslant s(x, y) \leqslant S_{(k)} \right\}.$$

Next, for the trivial case $k = n + 1$, we have $\mathcal{C}^{\text{mv}}(x) = \mathcal{C}_1(x) = \mathcal{C}_2(x) = \mathcal{Y}$. Therefore:

$$\mathcal{C}^{\text{mv}}(X_{n+1}) = \begin{cases} \left\{ y \in \mathcal{Y} \mid S_{(n+1-k)} \leqslant s(X_{n+1}, y) \leqslant S_{(k)} \right\} & k = \lceil (n+1)(1-\alpha) \rceil < n+1, \\ \mathcal{Y} & k = n+1. \end{cases}$$

A simple counting of the probability of placing S_{n+1} gives

$$P\left\{ Y_{n+1} \in \mathcal{C}^{\text{mv}}(X_{n+1}) \right\} = \frac{\max\{0, 2k - n - 1\}}{n+1} = \max\left\{ 0, 1 - \frac{2\lfloor \alpha(n+1) \rfloor}{n+1} \right\}.$$

It just happens that this formula works for $k = n + 1$ case as well. This completes the computation. ////